Geometric Feature Injection and Class-Balanced Sampling for Ink Detection on Herculaneum Papyri

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***Abstract*—Ink detection in micro-CT scans of carbonized Herculaneum papyri is a challenging inverse problem due to the low X-ray contrast between carbon ink and the papyrus substrate, compounded by extreme class imbalance ($< 2\%$ ink coverage). In this work (M3), we propose a novel improvement over the standard intensity-based 3D U-Net baseline established in Milestone 2. Our contribution is threefold: (i) we introduce geometric feature injection by augmenting the input with 3D surface normals $(I, \nabla_z I, \nabla_y I, \nabla_x I)$ to resolve intensity ambiguity; (ii) we implement strict class-balanced sampling to stabilize learning dynamics; and (iii) we utilize rotational augmentation to enforce orientation invariance. Evaluated on Fragment 1, our method improves the $F_{0.5}$ score from 0.6565 (baseline) to 0.7184, achieving a high Precision of 0.8343 and a PSNR of 32.05 dB. We also report a critical negative result regarding depth-roll augmentation, offering a robustness guideline for volumetric training under 2D supervision.**

***Index Terms*—Herculaneum Papyri, Ink Detection, 3D U-Net, Geometric Features, Volumetric Segmentation.**

I. INTRODUCTION

The Vesuvius Challenge provides high-resolution micro-CT volumes of ancient scrolls, where the primary objective is to segment ink patterns to recover lost text [3]. While virtual unwrapping techniques have successfully read scrolls written with metal-based inks (e.g., En-Gedi [1]), the Herculaneum papyri present a unique difficulty: the ink is carbon-based, making it nearly isodense with the carbonized papyrus substrate.

Parker et al. [2] overturned the assumption that these scrolls were unreadable by demonstrating that ink induces subtle micro-structural cues—such as surface smoothing and crack morphology—that are detectable via machine learning. However, standard baselines often struggle to learn these features due to:

1) **Intensity Ambiguity:** Compressed papyrus fibers often exhibit density profiles indistinguishable from ink.
2) **Extreme Imbalance:** The positive class (ink) is extremely rare, leading to gradient collapse.
3) **Directional Overfitting:** Models frequently memorize the orientation of fibers rather than the texture of the ink.

In our previous milestone (M2) [**?**], we established a baseline using a compact 3D U-Net trained on raw intensity. While reproducible, it suffered from high noise levels ($F_{0.5} = 0.6565$, Precision < 0.65).

Contribution: In this work (M3), we address these limitations by explicitly modeling the physical properties of the ink. We propose a Geometric-Aware pipeline that computes 3D gradients to highlight surface texture and enforces rotational invariance to suppress fiber noise.

II. RELATED WORK

A. State-of-the-Art Approaches

Current state-of-the-art methods typically rely on massive ensembles or specialized architectures. Quattrini et al. [4] introduced Volumetric Fast Fourier Convolution (vFFC) to mix local spatial features with global frequency reasoning, establishing a strong baseline. Other top-performing approaches in the Vesuvius Challenge utilize "2.5D" architectures (ResNet3D encoders with 2D decoders) to leverage pre-trained video recognition weights [5]. Unlike these resource-intensive methods, our work focuses on optimizing the *single-model* performance of a lightweight 3D U-Net, making it accessible for limited-resource settings while still achieving competitive results on Fragment 1.

B. Physical Basis for Detection

Our approach is grounded in the physical observations of Parker et al. [2], who noted that ink sits *on top* of the substrate, creating a measurable thickness change (10-20 microns). We hypothesize that explicit gradient calculation captures this morphology better than implicit feature extraction from raw intensity.

III. METHODOLOGY (OWN CONTRIBUTION)

We adopt the M2 baseline architecture (a 4-layer 3D U-Net with input depth $D = 4$) but fundamentally modify the input signal and training dynamics.

A. Contribution A: Geometric Feature Injection

Standard models rely on the scalar intensity field $I(x, y, z)$. We introduce a multi-channel input tensor to capture local surface geometry. We compute the gradient vector field ∇I using central finite differences:

$$\nabla I = \left[\frac{\partial I}{\partial z},\ \frac{\partial I}{\partial y},\ \frac{\partial I}{\partial x}\right] \tag{1}$$

These gradients are computed on the normalized volume and stacked with the original intensity, resulting in a 4-channel input $X \in \mathbb{R}^{4\times D\times H\times W}$. These channels approximate the surface normals, forcing the network to learn filters that respond to *shape changes* (the "crackle" of ink) rather than just brightness.

B. Contribution B: Strict Class-Balanced Sampling

The dataset is highly imbalanced ($\approx$ 98% background). In preliminary experiments, random sampling yielded batches with zero ink pixels, causing loss instability. We implement a custom **Balanced Sampler** that forces every batch to contain exactly 50% ink-centered patches and 50% background patches. This artificially inflates the prevalence of ink during training, ensuring consistent gradient updates for the positive class.

C. Contribution C: Rotational Invariance TTA

Papyrus fibers are anisotropic (directional), while ink texture is isotropic. To prevent overfitting to fiber direction, we apply:

- **Train-Time:** Random $k \cdot 90^\circ$ rotations and axis flips ($8\times$ data expansion).
- **Test-Time Augmentation (TTA):** We average predictions across three views: Original, Horizontal Flip, and Vertical Flip.

This acts as a regularizer: structures that look like ink only in one orientation (e.g., vertical fibers) are suppressed by the averaging process.

IV. Robustness Insight: The Failure of Z-Roll

We experimented with **Depth-Roll Augmentation** (shifting the volume along the z-axis) to improve robustness. **Observation:** Validation performance collapsed ($F_{0.5} < 0.20$), with the model predicting ink everywhere. **Explanation:** Our supervision is 2D (fixed to the surface mask). Rolling the volume moves the physical ink from the supervised center slice ($z = 2$) to an unsupervised slice ($z = 0$), while the label remains "Ink." This creates contradictory training pairs (empty input mapped to positive label). We conclude that Z-roll is incompatible with 2D-supervised volumetric training unless the loss function is aggregation-based (e.g., Max-Pooling).

V. Experimental Setup

Dataset: Fragment 1 (Vesuvius Challenge), split chronologically (Top 80% Train, Bottom 20% Validation). **Protocol:** Trained for 100 epochs on dual NVIDIA T4 GPUs using AdamW ($lr = 10^{-3}$) and BCEWithLogitsLoss. **Metrics:** The primary metric is $F_{0.5}$ (beta=0.5), prioritizing precision. We also report PSNR and Pseudo F-Measure (pFM) for structural analysis.

VI. Results

Table I summarizes the performance. Our M3 model significantly outperforms the M2 baseline.

TABLE I
Quantitative Comparison: Baseline (M2) vs. Proposed (M3)

Method	$F_{0.5}$	**Precision**	**Recall**	**pFM**	**PSNR**
M2 Baseline	0.6565	(Low)	**0.7241**	**0.5830**	11.56 dB
M3 (Ours)	**0.7184**	**0.8343**	0.4618	0.5626	**32.05 dB**

VII. Analysis and Discussion

A. Performance Gains

The M3 model achieves an $F_{0.5}$ of **0.7184** (+9.4% improvement). The primary driver is **Precision (0.8343)**. The baseline often confused dense fiber bundles with ink. By adding gradient channels, the M3 model successfully distinguished the "rounded" geometry of fibers from the "sharp" boundary of ink. The rotational invariance further filtered out directional false positives.

B. Trade-offs and Image Quality

While Recall decreased (0.46 vs 0.72), **PSNR improved drastically (+20 dB)**. This indicates a shift from a "noisy, high-recall" detector (M2) to a "clean, high-precision" specialist (M3). In digital papyrology, high precision is preferred to prevent the hallucination of non-existent characters. The slight drop in pFM (0.56 vs 0.58) reflects this conservative tuning, but the visual quality of the reconstruction is significantly higher.

VIII. Conclusion

We have successfully developed a Geometric-Aware ink detection model. By explicitly modeling surface normals and enforcing rotational invariance, we surpassed the baseline performance for this architecture. Our results confirm that for carbon-based ink detection, local surface geometry is a far more robust predictor than X-ray intensity alone.

References

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